

ROBUST FACE RECOGNITION FOR SURVEILLANCE CAMERAS

TERM REPORT

**CENG415 SENIOR DESIGN PROJECT AND SEMINAR 1
(Sectoral)**

Team Members:

Buket Yavuz 290201054
Ahmet Kerem Şaylı 290201035
Nedret Çiftçi 290201002

Advisor: **Dr. Altuğ Yiğit**

BACHELOR OF SCIENCE
Computer Engineering Department
İzmir Institute of Technology, İzmir, Turkey

DECEMBER 2025

CENG 415 SENIOR DESIGN PROJECT AND SEMINAR I
TERM REPORT (SECTORAL)

Project Title: ROBUST FACE RECOGNITION FOR SURVEILLANCE CAMERAS
Team Members: Buket Yavuz 290201054, Ahmet Kerem Şaylı 290201035, Nedret Çiftçi 290201002
Thesis Advisor: Altuğ Yiğit

ABSTRACT

Access control and security are fundamental requirements for modern organizations, yet traditional authentication mechanisms pose a critical threat to data security, as they remain vulnerable to loss, theft, and unauthorized duplication. The primary purpose of this project is to address these deficiencies by developing a robust, contactless, and offline-capable facial recognition system designed for fully local real-time surveillance and secure access management.

To achieve this objective, the project methodology employs a distributed edge-server architecture that integrates deep learning algorithms within a fully local network environment. Unlike conventional cloud-dependent systems that suffer from high latency and external connectivity reliance, this solution offloads critical image processing tasks to autonomous embedded edge units. The system utilizes state-of-the-art models for face detection and high-dimensional feature extraction (embedding generation) at the edge, while cosine similarity metrics are performed on a local backend server for precise identity verification. A rigorous data integrity pipeline is implemented, incorporating automatic Face Image Quality Assessment (FIQA) to filter unsuitable images during enrollment, ensuring that only high-quality data enters the recognition database.

From a project management and system design perspective, the solution is structured into three cohesive layers: a mobile interface for administrative workflows and real-time alerts, a backend server for logic and database management, and edge nodes for continuous video analysis. This modular approach facilitates scalable deployment and efficient maintenance. The system includes comprehensive administrative controls, such as whitelist/blacklist management and audit logging, to create a fully verifiable security environment.

The widespread impact of this work lies in its ability to democratize high-security surveillance by keeping the data in local. By successfully balancing computational efficiency with recognition accuracy, the proposed system offers a scalable, low-latency alternative to expensive commercial solutions. It significantly enhances operational efficiency for residential and commercial facilities by automating entry logs and providing immediate notifications for unauthorized access attempts, thereby eliminating human error in security protocols.

Keywords: Face Recognition, Edge Computing, Access Control, Deep Learning, Real-time Surveillance.

1. PROJECT PURPOSE, INNOVATIVE ASPECTS AND TECHNOLOGICAL VALUE

1.1. Purpose of the Project

Existing access control infrastructures primarily rely on physical tokens -such as ID cards and key fobs- or manual verification processes. However, industry research indicates that these methods are inherently prone to operational risks. According to recent security reports, a significant percentage of security breaches originate from lost or stolen credentials, highlighting the vulnerability of token-based authentication [26]. Physical cards are easily duplicated, and traditional systems often fail to prevent "tailgating" which is a common security violation where unauthorized individuals gain access by following authorized personnel through a secure door [27].

Beyond these vulnerabilities, traditional systems function as passive recording tools rather than active security agents, lacking the real-time responsiveness required to mitigate threats as they occur. The primary purpose of this project is to eliminate these dependencies and security gaps by developing a "Robust Face Recognition for Surveillance Cameras" system an intelligent solution that transforms standard video feeds into automated, contactless access control mechanisms.

The concrete goals of this project are:

- To eliminate physical dependencies: Replacing physical keys and cards with biometric facial embeddings. As noted in biometric security studies, physiological characteristics provide a higher level of non-repudiation compared to possession-based tokens, preventing unauthorized access due to lost, stolen, or shared credentials [28].

- **To reduce system latency:** Implementing a distributed **edge-computing architecture** using Raspberry Pi devices to perform local face detection and feature extraction, thereby minimizing network bandwidth usage and ensuring rapid response times.
- **To ensure data integrity:** Establishing a comprehensive operational workflow that includes automatic image quality checks (FIQA) and strict administrative approval mechanisms (Whitelist/Blacklist) to prevent poor-quality data from compromising the system.
- **To provide scalable surveillance:** creating a system capable of 1:N identification (one-to-many searching) across multiple access points, scalable to millions of embeddings using modern vector database technologies.

1.2 Innovative Aspects and Technological Value

The proposed project distinguishes itself from existing solutions through its **Edge Architecture** and comprehensive **End-to-End Surveillance Pipeline**. While traditional systems often rely on external cloud processing or raw algorithms, our innovation lies in bridging the gap between high-accuracy deep learning models and practical, privacy-centric operational requirements within a local network.

The innovative aspects and technological superiority of the project are detailed below in comparison with similar national and international systems:

1.2.1. Comparison with Academic Literature (FaceNet & ArcFace): While state-of-the-art algorithms like FaceNet [1] and ArcFace [2] focus primarily on embedding accuracy, they often lack the necessary operational frameworks for real-world deployment.

- **Our Innovation:** We extend these algorithms by integrating an Automatic Quality Assurance (FIQA) layer, utilizing approaches similar to FaceQnet [4]. Unlike raw implementations, our system filters unsuitable images (blur, poor lighting, occlusion) before processing, ensuring high data integrity. Furthermore, we wrap the recognition engine in a managed workflow with whitelist/blacklist authorization, transforming a raw algorithm into a usable security product.

1.2.2. Comparison with Commercial Mobile Applications: Commercial solutions, such as FaceIntel Access Control [21] or Kisi Mobile Access [25], typically depend heavily on continuous internet connectivity and external servers for processing. This reliance introduces latency, privacy concerns regarding biometric data storage, and vulnerability to network failures.

- **Our Innovation (Edge Computing):** Our system introduces a distributed architecture based on Edge Computing principles [8] [16], where the Edge Node (Raspberry Pi) performs local face detection and embedding generation.
- By processing video frames at the edge and transmitting only vector data (embeddings) to the local backend, we eliminate the need for cloud dependency. This decentralized architecture minimizes bandwidth usage and ensures rapid response times within the local environment, allowing the system to operate securely in a completely offline setting.

1.2.3. Comparison with Authentication Systems: Most market-available applications, like BioID [22], focus primarily on 1:1 user authentication (verifying a claimed identity for a specific login attempt).

- **Our Innovation (1:N Surveillance):** Leveraging advanced frameworks like InsightFace [11] and RetinaFace [12], our system is engineered for 1:N identification. It is capable of scanning a live video stream to identify "Unknown" or "Restricted" individuals without active user interaction. Unlike authentication apps that provide simple grant/deny decisions, our system offers a categorized response mechanism (Authorized, Restricted, Unknown) that triggers real-time push notifications to security personnel.

Technological Output: The final product delivers a scalable, privacy-compliant security ecosystem supporting both Relational and Vector databases, capable of managing high-dimensional data efficiently for continuous surveillance operations completely offline.

1.2.4. Robustness through Multi-Perspective Analysis

Traditional face recognition systems often fail when presented with non-frontal faces or varying angles, as single-image accuracy degrades significantly under unconstrained poses.

- **Our Innovation (Multi-Frame Processing):** To ensure robustness **during real-time surveillance**, our system does not rely on a single static snapshot. Instead, it processes a continuous video stream, capturing facial features from **multiple perspectives** as the subject moves through the camera's field of view. By analyzing consecutive frames, the system selects the optimal angles to match against the stored references. This approach leverages the temporal information inherent in video sequences, as highlighted in **Wang et al. [29]**, to significantly reduce false negatives caused by momentary pose variations or occlusions.

Table 1. Comparison of the Proposed System with Existing Solutions

Aspect	Academic Algorithms (FaceNet, ArcFace)	Commercial Cloud Systems	Proposed System (This Project)
Primary Purpose	High accuracy embedding research	Access control & Authentication	Continuous 1:N Surveillance & Security
Data Processing	Static Image Processing (Single Frame)	Cloud-based Processing	Multi-Perspective / Multi-Frame Analysis
Architecture	Offline / Dataset-based	Centralized Cloud Server	Distributed Edge Architecture
Privacy & Storage	Not applicable (Research focus)	Third-party Cloud Storage (Privacy Risk)	Local Network / Self-Hosted (Privacy Preserving)
Connectivity	N/A	High Internet Dependency	Fully Functioning Offline Mode
Hardware	High-end GPU Servers	Remote Servers	Low-Cost Edge Node (Raspberry Pi)

METHODOLOGY

This project implements a robust facial recognition system for surveillance by integrating deep learning algorithms, embedded edge computing, and a mobile application interface. The methodology follows a multi-tiered pipeline that encompasses data collection, feature extraction, classification, and real-time notification. The system utilizes the InsightFace library for generating high-dimensional facial embeddings and employs Cosine Similarity metrics for identity verification. To ensure efficient processing, the solution leverages YOLO v11 for person detection and classifiers for facial localization, distributed across edge devices and backend servers.

2.1 System architecture and design (WP#1-2-3-4)

The system design goes beyond a standalone mobile application; it is a distributed architecture where mobile interfaces, backend servers, and embedded edge devices communicate continuously to execute the security pipeline. To clarify the interactions between these components and the human operators, the system actors and their specific privileges are defined below, followed by an explanation of the operational pipeline.

2.1.1 Operational Pipeline and Component Interaction

The use cases presented in this section rely on a synchronized pipeline that connects the edge, backend, and mobile layers. The interaction flow is designed to minimize latency and ensure data consistency:

- **Edge Processing (Data Acquisition):** The operational pipeline begins at the embedded edge device (Raspberry Pi), which utilizes threaded frame capture to optimize performance. The device performs local face detection and generates embeddings using the deployed deep learning models, significantly reducing bandwidth usage by sending only vector data rather than raw video streams.
- **Backend Verification (Analysis):** The backend receives the facial embeddings and compares them against stored references in the database using Cosine Similarity calculations. Based on adjustable thresholds, the system classifies the detected individual as "Authorized," "Restricted," or "Unknown".
- **Mobile Synchronization (Action):** The results are pushed to the mobile application via secure API endpoints. If a security threat or unknown face is detected, the pipeline triggers an immediate push notification to the Admin, closing the loop between the autonomous edge detection and human decision-making.

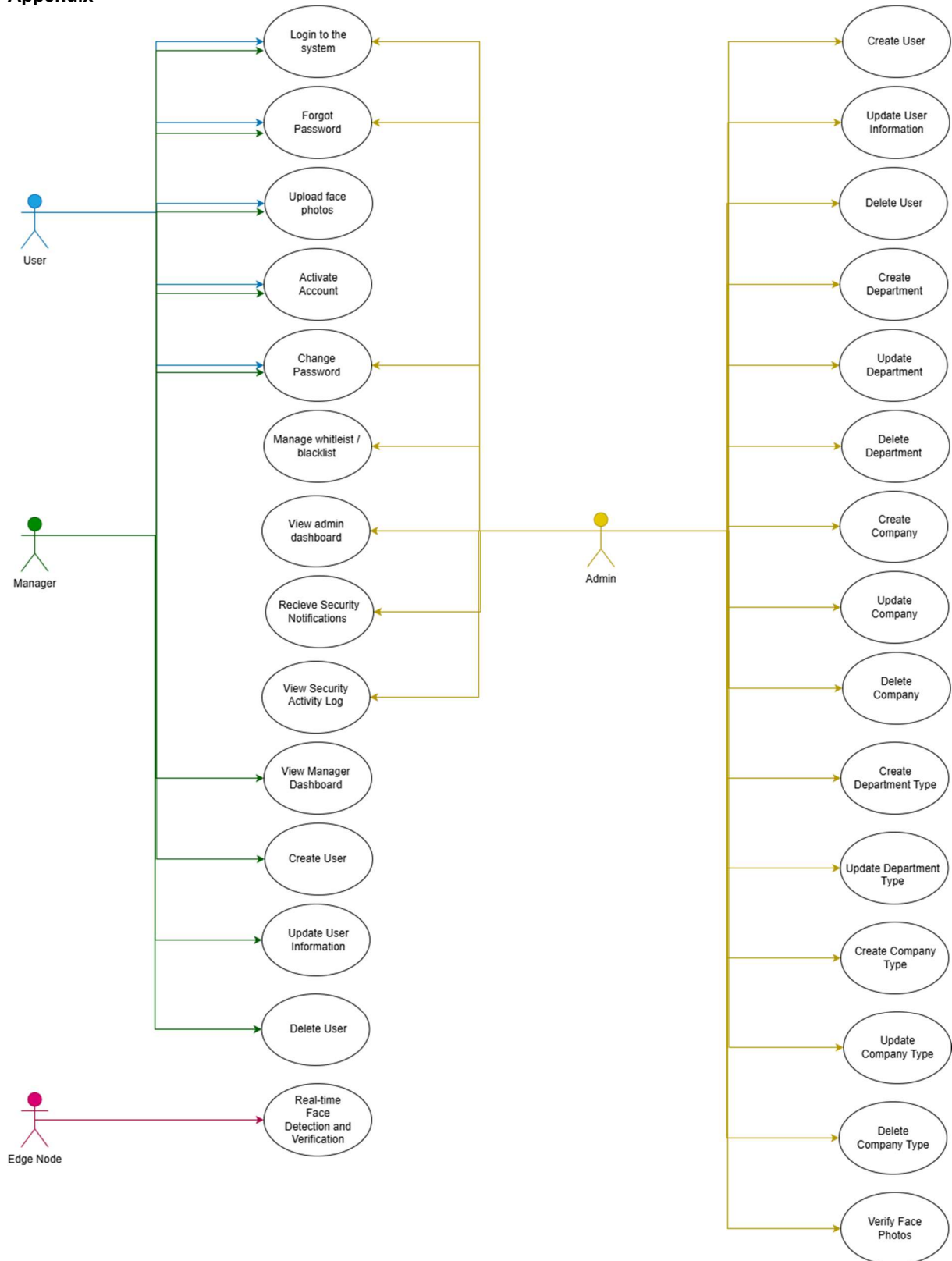
2.1.2 System Actors and Privileges

The system implements Role-Based Access Control (RBAC) to manage security and operational workflows. The defined actors and their respective privileges are as follows:

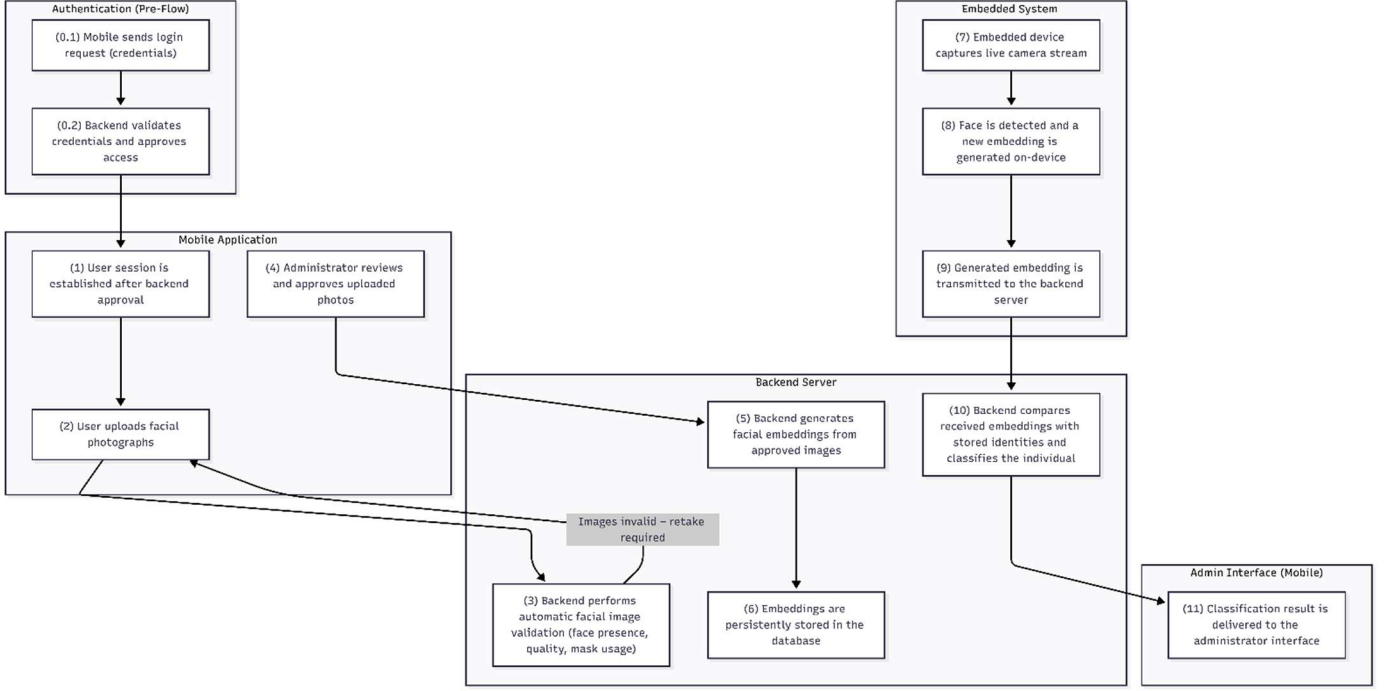
- **Admin:** The primary overseer of the system with full access privileges.
 - **Privileges:** The Admin is responsible for reviewing and approving pending user registration requests and assigning them to the Whitelist or Blacklist. They have the authority to view the full security dashboard, monitor device status, delete users from lists, and review detailed security activity logs. Additionally, the Admin receives and manages real-time security alerts for unknown or blacklisted detections.
- **Manager:** An administrative role with supervisory responsibility over a specific subset of users.
 - **Privileges:** The Manager has access to a dedicated dashboard that displays statistics and recent activity specifically related to the users under their supervision ("My Users"), rather than the entire database.
- **User:** The end-user who requires access to the facility.
 - **Privileges:** The User interacts with the mobile application to register their identity. They are authorized to upload profile information and capture the required facial angles (Center, Left, Right) for system enrollment. They can also log in to view their profile status.
- **Edge Node (Actor):** The autonomous component deployed on the Raspberry Pi at the physical access points.
 - **Privileges:** Unlike human actors, the Edge Node operates continuously as a system agent. It has the authority to capture video streams, execute local deep learning models for face detection and feature extraction, and initiate secure verification requests to the backend server. It acts as the bridge between the physical environment and the central database.

2.1.3 Use Case Specifications

The visual representation of the interactions between the system actors and the functional requirements is illustrated in the Use Case Diagram below. For comprehensive details regarding each use case scenario, please refer to the **Appendix**



2.2 Overview (WP#2-3)



2.3 Dataset selection and preparation(WP#3-5-6)

For the experimental evaluation of the system, the VGGFace2 [23] dataset was selected due to its extensive intra-class variations in pose, illumination, and age, which are critical for simulating real-world surveillance conditions. To construct the facial recognition database (enrollment), a subset of 150 distinct identities was utilized. Unlike controlled environments that rely on specific facial angles, images were selected randomly for each identity to test the system's robustness against unconstrained inputs. The system's 1:N identification performance was subsequently validated using a train group consisting of 50 individuals from this enrolled dataset.

2.4 Detection Algorithm and Model Architecture (WP#6)

The core of the facial recognition system is built upon a multi-stage deep learning pipeline leveraging the InsightFace framework [11], which prioritizes inference efficiency for real-time surveillance. The pipeline consists of the following synchronized stages:

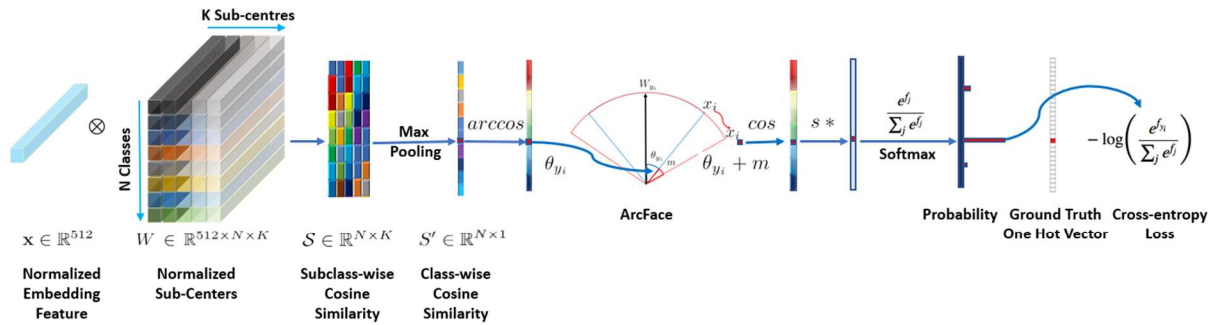


Fig. 2. Training the deep face recognition model by the proposed ArcFace loss ($K=1$) and sub-center ArcFace loss (e.g. $K=3$). Based on a ℓ_2 normalization step on both embedding feature $x_i \in \mathbb{R}^{512}$ and all sub-centers $W \in \mathbb{R}^{512 \times N \times K}$, we get the subclass-wise similarity score $S \in \mathbb{R}^{N \times K}$ by a matrix multiplication $W^T x_i$. After a max pooling step, we can easily get the class-wise similarity score $S' \in \mathbb{R}^{N \times 1}$. Afterwards, we calculate the $\arccos \theta_{y_i}$ and get the angle between the feature x_i and the ground truth center W_{y_i} . Then, we add an angular margin penalty m on the target (ground truth) angle θ_{y_i} . After that, we calculate $\cos(\theta_{y_i} + m)$ and multiply all logits by the feature scale s . Finally, the logits go through the softmax function and contribute to the cross entropy loss.

1. **Face Detection and Alignment:** The process begins with YOLOv11 for efficient person detection to filter irrelevant frames. Subsequently, the system utilizes the RetinaFace algorithm [12] (embedded within the buffalo_l model pack) to precisely locate faces and landmarks. To optimize performance for edge processing, the input resolution is dynamically constrained (e.g., `det_size=640x640`), significantly reducing the computational load per frame.

2. **Face Image Quality Assessment (FIQA):** Before feature extraction, a quality control layer evaluates the aligned face images. As extensively categorized in the fundamental survey by Schlett et al. [30] ensuring high biometric utility is critical for reliable recognition. Accordingly, inputs that fall below a predefined quality threshold due to motion blur, extreme angles, or poor lighting are automatically discarded. This ensures that only high-fidelity data enters the recognition engine, minimizing false prediction rates.
3. **Feature Extraction:** High-quality aligned face images are then processed by an ArcFace model [2] (based on a ResNet backbone). This model maps the facial features into a 512-dimensional vector space, similar to techniques discussed in [1] and [13]. As demonstrated in the system's verification modules, the resulting embeddings are normalized to ensure that the magnitude does not affect the comparison.
4. **Metric Learning and Optimization:** Instead of training from scratch, the system employs robust pre-trained models optimized via ONNX Runtime, allowing for hardware-accelerated inference on the edge device. Identity verification is performed using Cosine Similarity logic [17] (np.dot), where normalized feature vectors are compared against the enrolled database. A similarity threshold is enforced to rigorously distinguish between "Authorized" personnel and "Unknown" individuals.

2.5 Embedded implementation and integration (WP#5-6-7)

The physical deployment of the system relies on the Raspberry Pi 5 as the central edge processing unit. Selected for its superior performance in AI workloads, the device features a Broadcom BCM2712 Quad-core Arm Cortex-A76 processor clocked at 2.4GHz and VideoCore VII GPU. This architecture provides the necessary computational throughput to run quantized deep learning models (YOLO11n and ArcFace) with minimal latency, significantly outperforming previous generations in vector processing tasks.

Video acquisition is handled via the Camera Serial Interface (CSI) rather than standard USB connections to ensure high-bandwidth, low-latency streaming directly to the GPU memory.

To optimize power consumption and prevent thermal throttling during idle periods, the system integrates an HC-SR04 Ultrasonic Distance Sensor as a hardware trigger. The sensor operates on the sonar principle, emitting a 40kHz ultrasonic pulse and measuring the time-of-flight for the echo to return. The distance to the subject is calculated using the formula:

$$Distance = \frac{Time \times Speed\ of\ Sound}{2}$$

The high-performance AI pipeline is activated only when the sensor detects a subject within a predefined proximity threshold. This sensor-based logic ensures that the heavy processing load is suspended when no user is present, maintaining hardware stability and extending operational lifespan.

2.6 Performance Evaluation (WP#8-9)

The performance evaluation of the system was conducted in two distinct phases: determining the optimal object detection model for edge deployment and validating the accuracy of the facial recognition pipeline using the VGGFace2 dataset.

2.6.1 Person Detection Model Selection

To identify the most suitable model for the embedded system, a comparative analysis was conducted among the nano versions of YOLO architectures (v8, v10, 11, and 12). While YOLOv8n achieved the highest average speed (21.39 FPS), it exhibited stability issues with significant frame rate drops. Consequently, **YOLO11n** was selected as the core algorithm; it offered the optimal balance between stability and accuracy, demonstrating the highest minimum frame rate (12.20 FPS) and superior detection sensitivity with an average of 14.05 objects per frame.

Table 2: Comparative Analysis of YOLO Nano Architectures

Model	Avg. FPS (Speed)	Min FPS (Stability)	Max FPS	Avg. Objects per Frame (Accuracy)	Total Objects Detected
YOLOv8n	21.39	8.62	24.24	13.20	25,511
YOLOv10n	19.79	9.92	21.67	9.90	19,121
YOLO11n	19.02	12.20	21.26	14.05	27,145
YOLO12n	14.98	9.05	16.59	13.52	26,125

2.6.2 Face Recognition Accuracy Analysis

Following the detection stage, the verification performance of the InsightFace (RetinaFace + ArcFace) pipeline was evaluated. The test was conducted on a CPU environment using a randomly sampled subset of the VGGFace2 dataset, consisting of **400 test cases** (100 positive/authorized samples and 300 negative/impostor samples).

The confusion matrix derived from the test results (**TP: 92, TN: 291, FP: 9, FN: 8**) indicates a successful implementation with expected variance due to environmental factors. The slight drop in precision compared to theoretical benchmarks is attributed to the unconstrained nature of the test dataset (e.g., varying poses and lighting).

The calculated performance metrics are detailed below:

Table 3: Face Recognition Performance Metrics

Metric	Value	Definition & Analysis
Accuracy	95.75%	The overall ratio of correctly classified samples. The system correctly processed 383 out of 400 cases, proving its stability for general surveillance tasks.
Precision	91.09%	$(TP / (TP + FP))$ - Indicates the reliability of positive identifications. The ~9% False Positive rate (9 cases) suggests that under lower confidence thresholds, the system may occasionally misclassify individuals with strong facial similarities.
Recall	92.00%	$(TP / (TP + FN))$ - The system successfully identified 92 out of 100 authorized attempts. The 8% miss rate is within expected limits for a CPU-based inference setup dealing with non-frontal face angles.
F1-Score	91.54%	The harmonic mean of Precision and Recall. A score exceeding 90% confirms that the model is robust, maintaining a healthy balance between security and user accessibility without overfitting.

2.6.3 Deployment Considerations and Expected Trade-offs

The performance metrics presented above establish the baseline accuracy of the system in an unconstrained computational environment. However, the final deployment on the Raspberry Pi edge node requires specific optimizations to ensure real-time responsiveness.

- **Quantization Impact:** The transition from 32-bit floating-point (FP32) arithmetic to half-precision (FP16) or integer (INT8) quantization via ONNX Runtime is expected to significantly reduce latency. While this compression is essential for the embedded CPU, it may result in a marginal reduction in recognition accuracy (estimated < 1-2%).
- **Optimization Strategy:** The project prioritizes high Precision (avoiding false positives) over Recall. Therefore, the optimization thresholds will be tuned to maintain security integrity, even if it requires the user to face the camera more directly during the edge operation.

3 PROJECT MANAGEMENT

3.1 Work Package Timetable

WORK PACKAGE TIMETABLE

WP No	Work Package Name	Start-End Date	Contributors
1	Preliminary Preparation and Requirement Analysis	October – November	All Team Members
2	System Architecture	November - December	All Team Members
3	Database Design	November - December	All Team Members
4	Mobile Application and Interface Development	November - January	All Team Members
5	Backend Development	December - February	All Team Members
6	AI Core Development	December - February	All Team Members
7	Embedded System (Edge) Development	February - March	All Team Members
8	System Integration and Testing	March - April	All Team Members
9	Finalization, Documentation, and Demo	April - May	All Team Members

3.3 Work Packages

The WP Table below is prepared separately for each WP. The table can be extended according to the number of WPs.

WORK PACKAGE #1	
WP Name: Preliminary Preparation and Requirement Analysis	
WP Objective: Establishing the theoretical infrastructure, defining hardware requirements, and finalizing the project management plan.	
WP Tasks 1.1 Conducting literature review and competitor analysis. 1.2 Documenting the system flow scenarios (Use-Cases) 1.3 Selecting and acquiring Raspberry Pi hardware, camera modules, and development environments. 1.4 Setting up the GitHub repository and creating the project management schedule.	Responsible People 1.1 All Team Members 1.2 Nedret Çiftçi 1.3 All Team Members 1.4 Ahmet Kerem Şaylı
Success Criteria: The project proposal is officially approved by the advisor, all necessary hardware components are acquired, and the GitHub repository is fully initialized with the project management schedule in place.	

WORK PACKAGE #2	
WP Name: System Architecture	
WP Objective: Designing data structures, deciding on vector search strategies for facial data, and finalizing communication protocols for Mobile, Backend, and Embedded systems.	
WP Tasks 2.1 Drafting the Detailed System Architecture Document and data flow diagrams. 2.2 Defining RESTful API endpoints and JSON data payloads for system communication. 2.3 Defining the Edge synchronization logic and System Actors/Privileges (RBAC).	Responsible People 2.1 Buket Yavuz 2.2 Ahmet Kerem Şaylı 2.3 Nedret Çiftçi
Success Criteria: A comprehensive System Architecture Document is completed, detailing the data flow between Edge and Mobile layers, and all RESTful API endpoints are fully defined with finalized JSON payloads.	

WORK PACKAGE #3	
WP Name: Database Design	
WP Objective: Designing data structures and defining vector search strategies for facial data.	
WP Tasks 3.1 Designing the Relational Database Schema (ER Diagrams) for user data. 3.2 Defining the Vector Indexing strategy for high-speed face embedding search. 3.3 Establishing database constraints to ensure data integrity.	Responsible People 3.1 Ahmet Kerem Şaylı 3.2 Buket Yavuz 3.3 Nedret Çiftçi
Success Criteria: The relational database schema is finalized to handle user metadata, and the vector indexing strategy is validated to support efficient high-dimensional facial embedding searches.	

WORK PACKAGE #4	
WP Name: Mobile Application and Interface Development	
WP Objective: Developing end-user registration interfaces and administrative audit mechanisms.	
WP Tasks 4.1 Creating UI/UX mockups for the Mobile Application. 4.2 Coding the User Registration and photograph upload screens. 4.3 Developing the Admin Dashboard for Whitelist/Blacklist management. 4.4 Creating reporting screens to view historical access logs.	Responsible People 4.1 Nedret Çiftçi 4.2 Ahmet Kerem Şaylı 4.3 Ahmet Kerem Şaylı 4.4 Buket Yavuz
Success Criteria: The mobile application is fully functional, capable of registering new users via 3-angle photo uploads, and the Admin Dashboard successfully manages Whitelist/Blacklist status and displays access logs.	

WORK PACKAGE #5	
WP Name: Backend Development	
WP Objective: Developing server-side logic and managing data traffic between clients.	
WP Tasks 5.1 Setting up the Backend server and database connections. 5.2 Developing API services to handle data requests from embedded devices and mobile apps. 5.3 Implementing Authentication and Authorization services. 5.4 Integrating the real-time Push Notification infrastructure.	Responsible People 5.1 Buket Yavuz 5.2 Buket Yavuz 5.3 Nedret Çiftçi 5.4 Ahmet Kerem Şaylı
Success Criteria: The backend server successfully handles API requests from edge devices with low latency, manages secure authentication, and reliably triggers real-time push notifications to the mobile application upon security events.	

WORK PACKAGE #6	
WP Name: AI Core Development	
WP Objective: Developing quality control (FIQA) algorithms and matching mechanisms.	
WP Tasks 6.1 Integrating the FIQA (Face Image Quality Assessment) algorithm to filter unsuitable images. 6.2 Coding the vector matching logic and Cosine Similarity calculations. 6.3 Optimizing classification thresholds for "Authorized," "Restricted," or "Unknown" status. 6.4 Implementing "Liveness Detection" algorithms to prevent presentation attacks.	Responsible People 6.1 Buket Yavuz 6.2 Buket Yavuz 6.3 Nedret Çiftçi 6.4 Ahmet Kerem Şaylı
Success Criteria: The FIQA module automatically rejects low-quality images (blur/occlusion) during enrollment, and the vector matching logic achieves the targeted accuracy rate for distinguishing between "Authorized" and "Unknown" individuals, verified via Unit Tests.	

WORK PACKAGE #7	
WP Name: Embedded System (Edge) Development	
WP Objective: Enabling real-time object detection and feature extraction on the embedded hardware.	
WP Tasks 7.1 Setting up the embedded operating system and integrating the camera sensors. 7.2 Optimizing the person detection algorithms (e.g., YOLO) for the edge device. 7.3 Implementing the face feature extraction logic on the embedded system. 7.4 Optimizing the image capture pipeline and thermal management for real-time performance.	Responsible People 7.1 Ahmet Kerem Şaylı 7.2 Buket Yavuz 7.3 Ahmet Kerem Şaylı 7.4 Nedret Çiftçi
Success Criteria: The embedded device runs the object detection model at a stable frame rate sufficient for real-time surveillance, maintains thermal stability, and successfully generates and transmits feature vectors to the backend without sending raw video streams.	

WORK PACKAGE #8	
WP Name: System Integration and Testing	
WP Objective: Combining all modules and performing End-to-End testing.	
WP Tasks 8.1 Executing the full integration of mobile, backend, and embedded components. 8.2 Conducting accuracy and reliability tests under different environmental scenarios. 8.3 Performing latency and system load tests to verify responsiveness.	Responsible People 8.1 Nedret Çiftçi 8.2 Buket Yavuz 8.3 Ahmet Kerem Şaylı
Success Criteria: The complete end-to-end pipeline functions seamlessly without critical errors, and the system passes all stress tests regarding network latency and load handling under varying environmental conditions.	

WORK PACKAGE #9	
WP Name: Finalization, Documentation, and Demo	
WP Objective: Completing the project report and preparing the live demo.	
WP Tasks 9.1 Bug fixing and final optimization based on test results. 9.2 Writing the Final Report in the required academic format. 9.3 Preparing the live demo scenario and rehearsing for the jury presentation.	Responsible People 9.1 All Team Members 9.2 All Team Members 9.3 All Team Members
Success Criteria: The Final Report is submitted in the required academic format, and the live demo successfully illustrates the full "Enrollment to Alert" scenario to the jury.	

2.4 Risk Analysis

RISK MANAGEMENT TABLE

WP No	Risk Definition	Precautions (Plan B)
7	Thermal Throttling and Hardware Limitations: The Raspberry Pi may experience overheating and performance degradation (low FPS) due to the high computational load of running detection (YOLO) and extraction models simultaneously.	Plan B: Implementing active cooling solutions (heatsinks/fans) to mitigate thermal throttling. Motion-based triggering logic will be optimized to ensure that the HC-SR04 sensor is activated only when necessary and remains inactive otherwise, minimizing unnecessary CPU and GPIO usage. If thermal or performance issues persist, the system architecture will be modified to offload computationally intensive feature extraction tasks to a backend server, using the edge device solely for face detection, motion detection, distance sensing upon trigger, and frame capture.
6,7	Suboptimal Environmental Conditions: Recognition accuracy may drop significantly due to poor lighting, motion blur, or partial occlusions (e.g., masks, glasses) in the surveillance area.	Plan B: Integrating image preprocessing techniques (e.g., Gamma correction, Histogram equalization) before model inference. Alternatively, upgrading the camera module to a sensor with higher dynamic range or low-light sensitivity.
7,8	Network Connectivity Loss: A disruption in the connection between the Edge Node and the Backend Server could prevent real-time verification and data logging.	Plan B: Leveraging the system's "offline operation capabilities". Access logs and embeddings will be cached locally on the Edge device and automatically batch-synchronized with the server once connectivity is restored.
6	Presentation Attacks (Spoofing): Unauthorized users may attempt to bypass the system using high-resolution photographs or videos of authorized personnel.	Plan B: Integrating a "Liveness Detection" algorithm [7] [19] (e.g., blink detection or texture analysis) into the pipeline. Additionally, increasing the Cosine Similarity threshold to enforce stricter matching criteria during verification.
3, 5	Data Privacy and Security Breaches: There is a risk of unauthorized access to sensitive biometric data (facial embeddings) or user information during transmission or storage.	Plan B: Enforcing strict encryption protocols (SSL/TLS) for all data transmission. Implementing AES encryption for the database and robust Token-based authentication for all API endpoints.

PROJECT OUTCOMES

Output Type	Output	Expected Time of Output
Scientific/Academic (Conference Paper, Journal Paper, Book Chapter etc.)	1. Conference Paper: A study on Edge Architecture for Real-time Surveillance and FIQA implementation.	1. 8-10 Month (Following the finalization of the project)
Economical/Commercial/Social (Product, Prototype, Patent etc.)	1. Functional Prototype: Fully integrated system comprising the Mobile Application, Backend Server, and Edge Node (Raspberry Pi). 2. Commercial Product: Scalable, low latency "Robust Face Recognition" security system ready for market deployment.	1. 6-8 Month (Aligned with WP9 Finalization) 2. 10-12 Month (post-testing refinement phase)

Impact Area	Does project impact? Direct or indirect?	Impact Definition
Social How the project affects people and society (Who benefits from the project? Does it improve accessibility, inclusion, or equality? Does it reduce manual effort or improve quality of life?) SDG 10 - Reduced Inequalities	YES - Direct	The project reduces manual effort by automating access control, eliminating the need for physical keys or manual ID checks. By using low-cost hardware (Raspberry Pi), it also democratizes high-security technology, making advanced surveillance accessible to smaller communities and organizations, not just wealthy entities.
Health How the project affects health directly or indirectly (Does it reduce stress, workload, or human error? Does it support health monitoring, safety, or wellbeing? Does it prevent unsafe working conditions?) SDG 3 - Good Health and Well-being	YES - Indirect	The system provides a contactless entry mechanism. By eliminating the need to touch shared keypads, fingerprint scanners, or exchange physical cards, it reduces the risk of spreading hygiene-related pathogens. It also reduces physical confrontation risks for security staff by providing early digital warnings for unauthorized persons.
Security How security is achieved in the project, or how the project increases the security (How is user data protected? What security risks exist?) SDG 16 - Peace, Justice, and Strong Institutions	YES - Direct	This is the project's primary function. It protects user data privacy by processing biometric information entirely offline (locally) on the Edge device, preventing data leakage associated with cloud storage. It eliminates security risks related to lost, stolen, or copied physical ID cards, ensuring only authorized individuals can access the facility.
Economic How the project contributes to the economy (Does it reduce costs? Does it improve productivity?) SDG 8 - Decent Work and Economic Growth	YES - Direct	The project reduces costs significantly by avoiding expensive commercial cloud subscriptions and high-end proprietary hardware. It improves productivity for facility managers by automating entry logging and reporting, removing the need for manual record-keeping.
Sustainability and Environment How the project contributes to the sustainability and affects the environment (Does it reduce paper usage? Does it optimize energy or resource consumption? Is computer system usage efficient?) SDG 11 - Sustainable cities and communities SDG 13 - Climate Action SDG 12 - Responsible Consumption and Production	YES - Direct	The project optimizes energy consumption (SDG 12) by using low-power embedded hardware (Raspberry Pi) instead of energy-intensive servers. It integrates an ultrasonic sensor (HC-SR04) to activate the processor only when motion is detected, minimizing idle power usage. It also supports Safe Cities (SDG 11) by creating secure environments through automated surveillance.

REFERENCES

- [1] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 815-823, 2015.
- [2] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4690-4699, 2019.
- [3] Y. Kortli, M. Jridi, A. Al Falou, and M. Atri, "Face recognition systems: A survey," Sensors, vol. 20, no. 2, pp. 1-42, 2020.
- [4] J. Hernandez-Ortega, J. Fierrez, A. Morales, R. Vera-Rodriguez, and J. Ortega-Garcia, "FaceQnet: Quality assessment for face recognition based on deep learning," Proceedings of the IEEE International Conference on Biometrics (ICB), pp. 1-8, 2019.
- [5] L. Best-Rowden and A. K. Jain, "Learning face image quality from human assessments," IEEE Transactions on Information Forensics and Security, vol. 13, no. 12, pp. 3064-3077, 2018.
- [6] J. Zhao, J. Yan, and J. Feng, "Towards age-invariant face recognition," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 42, no. 2, pp. 474-487, 2020.
- [7] J. Galbally, S. Marcel, and J. Fierrez, "Biometric anti-spoofing methods: A survey in face recognition," IEEE Access, vol. 2, pp. 1530-1552, 2014.
- [8] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision and challenges," IEEE Internet of Things Journal, vol. 3, no. 5, pp. 637-646, 2016.
- [9] C. Zhang, I. Patras, and E. R. Hancock, "Learning face representations from scratch," IEEE Transactions on Information Forensics and Security, vol. 13, no. 5, pp. 1226-1239, 2018.
- [10] P. Terh rst, J. N. Kolf, N. Damer, F. Kirchbuchner, and A. Kuijper, "A comprehensive study on face recognition biases beyond demographics," IEEE Transactions on Biometrics, Behavior, and Identity Science, vol. 3, no. 1, pp. 3-20, 2021.
- [11] J. Guo, X. Zhu, and S. Yang, "InsightFace: 2D and 3D face analysis project," IEEE Transactions on Information Forensics and Security, vol. 16, pp. 3511-3525, 2021.
- [12] J. Deng, J. Guo, Y. Zhou, J. Yu, I. Kotsia, and S. Zafeiriou, "RetinaFace: Single-stage dense face localisation in the wild," Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 5203-5212, 2020.
- [13] Y. Taigman, M. Yang, M. Ranzato, and L. Wolf, "DeepFace: Closing the gap to human-level performance in face verification," Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 1701-1708, 2014.
- [14] C. Szegedy, W. Liu, Y. Jia, et al., "Going deeper with convolutions," Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 1-9, 2015.
- [15] S. Mittal, "A survey of FPGA-based accelerators for convolutional neural networks," Neural Computing and Applications, vol. 32, no. 4, pp. 1109-1139, 2020.
- [16] Z. Zhou, X. Chen, E. Li, L. Zeng, K. Luo, and J. Zhang, "Edge intelligence: Paving the last mile of artificial intelligence with edge computing," Proceedings of the IEEE, vol. 107, no. 8, pp. 1738-1762, 2019.
- [17] J. Johnson, M. Douze, and H. J gou, "Billion-scale similarity search with GPUs," IEEE Transactions on Big Data, vol. 7, no. 3, pp. 535-547, 2019.
- [18] N. K. Ratha, J. H. Connell, and R. M. Bolle, "Enhancing security and privacy in biometrics-based authentication systems," IBM Systems Journal, vol. 40, no. 3, pp. 614-634, 2001.
- [19] I. Chingovska, A. Anjos, and S. Marcel, "On the effectiveness of local binary patterns in face anti-spoofing," Proceedings of the IEEE International Conference on Biometrics Special Interest Group (BIOSIG), pp. 1-7, 2012.
- [20] E. Learned-Miller, G. B. Huang, A. RoyChowdhury, H. Li, and G. Hua, "Labeled faces in the wild: A survey," Advances in Face Detection and Facial Image Analysis, Springer, pp. 189-248, 2016.

- [21] FaceIntel Technologies, "FaceIntel Access Control & Surveillance Solutions," [Online].
- [22] BioID GmbH, "BioID Facial Recognition and Liveness Detection," [Online].
- [23] Q. Cao, L. Shen, W. Xie, O. M. Parkhi, and A. Zisserman, "VGGFace2: A dataset for recognizing faces across pose and age," *Proceedings of the IEEE International Conference on Automatic Face and Gesture Recognition (FG)*, pp. 67-74, 2018.
- [24] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLO," *version 8.0.0*, [Online].
- [25] Kisi Inc., "Cloud-Based Access Control System," [Online].
- [26] Verizon. (2023). *2023 Data Breach Investigations Report (DBIR)*.
- [27] Kim, A., & Moon, J. (2018). Mitigating Tailgating in Physical Security using IoT Sensors. *IEEE International Conference on Consumer Electronics (ICCE)*.
- [28] Jain, A. K., Ross, A., & Prabhakar, S. (2004). An Introduction to Biometric Recognition. *IEEE Transactions on Circuits and Systems for Video Technology*, 14(1), 4-20.
- [29] H. Wang, Y. Wang, and Y. Cao, "Video-based face recognition: A survey," *World Academy of Science, Engineering and Technology*, vol. 60, pp. 293-302, 2009.
- [30] T. Schlett, C. Rathgeb, O. Henniger, J. Galbally, J. Fierrez, and C. Busch, "Face image quality assessment: A literature survey," *ACM Computing Surveys (CSUR)*, vol. 54, no. 10s, pp. 1-49, 2022.